

ECE 0301

ECE Problem Solving Using C++

Yang Liu

Fall 2025

Instructor Information

Instructor: Prof. Yang Liu (刘扬)

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12:30 PM - 1:50 PM and 4:30 PM - 6:00 PM

or by appointment

TA Information

Section 1 for Tuesday class

TA: Xinyang Wu (吴欣旻)

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TA Information

Section 2 for Thursday class

TA: Yaoran Yue (岳耀冉)

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QQ: 403082964

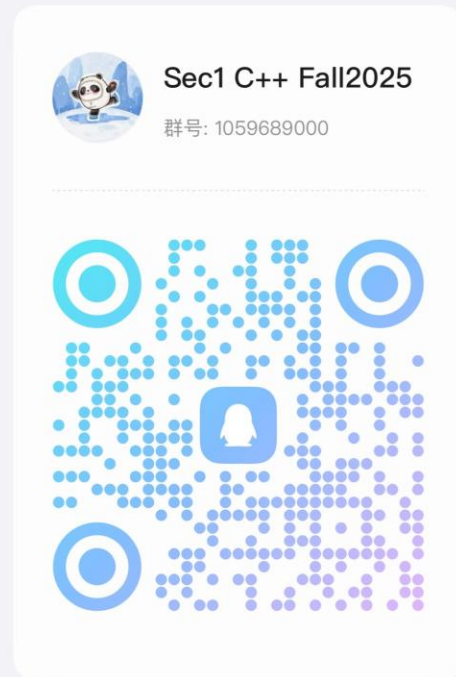
Course Information

Section 1

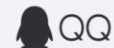
QQ Group: 1059689000

Homework assignment and answer, as well as important course announcement will be made in this QQ group.

It is very important for you to join this group.



扫一扫二维码，加入群聊



Course Information

Section 2

QQ Group: 903543357

Homework assignment and answer, as well as important course announcement will be made in this QQ group.

It is very important for you to join this group.



QQ Group Resource



群在线文档

可以与群成员同时在线编辑的文档



本周

按最新时间



RedPanda.C++.3.3.win64.MinGW64_11.5.0.Setup.exe

59.4 MB 16次

09-09

来自: 刘老师

2025-09



Starting Out with C++ - From Control Structures through Object...9).pdf

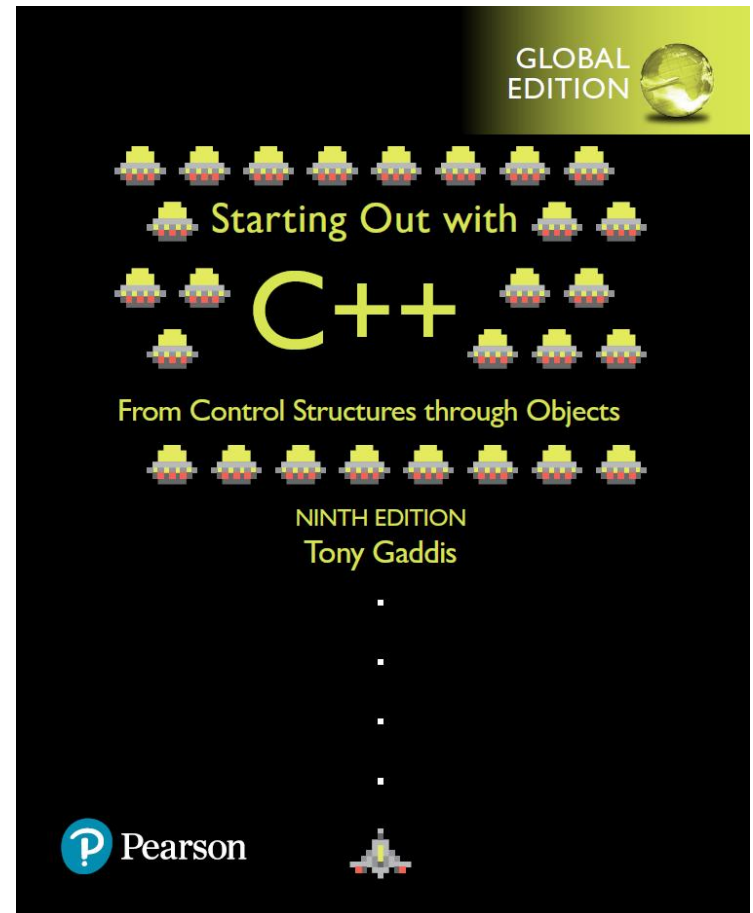
12.5 MB 12次

09-07

来自: 刘老师

Textbook

Tony Gaddis,
*Starting Out with C++
from Control Structures
through Objects*,
9th Edition, Pearson



About the author



- Author Tony Gaddis is the principal author of the **Starting Out With** series of textbooks. He has 20 years of experience teaching computer science courses, primarily at Haywood Community College.
- Tony is a highly acclaimed instructor who was previously selected as the North Carolina Community College "Teacher of the Year" and has received the Teaching Excellence award from the National Institute for Staff and Organizational Development. The Starting Out With series includes introductory textbooks covering programming logic and design, C++, Java(tm), Microsoft® Visual Basic®, Microsoft® Visual C#, Python, Alice and App Inventor, all published by Pearson.

Textbook on Amazon

Books › Computers & Technology › Programming Languages



Starting Out with C++ from Control Structures to Objects 9th Edition



by Tony Gaddis (Author)

4.4 ★★★★★ 572 ratings

[See all formats and editions](#)

Kindle
\$20.00

Read with Our **Free App**

Paperback
\$99.98

36 Used from \$61.75
3 New from \$218.72

Loose Leaf
\$138.48

4 Used from \$138.48
2 New from \$196.48

For two-semester courses in the C++ programming sequence, or an accelerated one-semester course.

A clear and student-friendly way to teach the fundamentals of C++

Starting Out with C++: From Control Structures through Objects covers control structures, functions, arrays,

and pointers before objects and classes in Tony Gaddis's hallmark accessible, step-by-step presentation. His books help beginning

✓ [Read more](#)



[See all 2 images](#)

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ISBN-10



9780134498379

ISBN-13



978-0134498379

Edition

#

9th

Publisher



Pearson

Publication date



February 13, 2017

Language



English



Textbook

Top reviews from the United States



vaaesthete

★★★★☆ **Appropriate for a CIS intro class to C++**

Reviewed in the United States on June 3, 2023

Verified Purchase

The material is presented at a basic, approachable level, with no prerequisites for programming. This is appropriate for CIS students and introductory C++ classes at the junior college level. You can also use this for self-study as the content is clearly explained and examples are amply provided. I would have preferred the authors committed to 'modern' C++ constructs (e.g., use C++11 initialization syntax instead of legacy C-like initialization) but otherwise the content is pretty good. The coding problems at the back of the chapters are pretty straightforward and are progressively more involved.

In terms of the actual Kindle version, the book was photocopied, with no attempt to make an actual digital edition of this book. As other have pointed out, several of the pages are upside-down and the links are not really usable to navigate.

The inclusion of recursion, dequeues, and trees seem a little out of place, but perhaps some classes cover this as a 2nd-semester topic.

Grading

Exam 1 20%

Exam 2 20%

Final Exam 30%

Homework 20%

Quiz 10%

The exam will be open book, open note, but no electronic devices are allowed except the calculator.

Letter Grade

Level	Letter Grade	Reported Numerical Score	Grade Points
Superior Performance	A	90 - 100	4.0
	A-	85 - 89	3.7
Meritorious Performance	B+	80 - 84	3.3
	B	76 - 79	3.0
	B-	73 - 75	2.7
Adequate Performance	C+	70 - 72	2.3
	C	66 - 69	2.0
	C-	63 - 65	1.7
Minimal Performance	D+	61 - 62	1.3
	D	60	1.0
Insufficient Performance (Failure)	F	< 60	0.0

Homework

- Homework problems will be assigned periodically and are due as stated. (HW questions will be uploaded to QQ group)
- **Late** submission will **NOT** be accepted.
- For example, if the deadline is on Monday (Wednesday), and you submit your homework at 12:01am on Tuesday (Thursday), it will be considered as late submission.

Homework

Submission Format

- **The homework should be submitted as a single PDF document.**
- **The homework can be done by typing in word and converting it into PDF.**
- **If the homework is handwritten, please write everything clearly.**
 - If it is written on paper, please take photos and merge them into a PDF document. You may need software like 扫描全能王 for merging.
 - If it is written on an iPad, such as Notability and GoodNotes, it can generate a PDF file directly.

Homework

- **Naming Convention**

**HW Number-last four digits of Student ID.
For example HW1-0193.**

- **Grading Criteria**

Every homework is 100 points.

The homework is graded based on honest efforts, especially completeness.

Homework

Submission Platform

Every homework should be submitted through **Blackboard**.

Before the deadline, the homework can be submitted multiple times and only the last version submitted will be graded by TA.

Homework

Homework Deadline

- It is usually due at 11:59 pm on the due date.
- Please pay attention to the deadline of each homework carefully.
- If you feel that you need an extension, please **contact the TA** before the homework deadline.

Avoiding Plagiarism

Unacknowledged direct copying from the work of another person, or the close paraphrasing of somebody else's work, is called plagiarism and is a serious offence, equated with cheating in examinations. This applies to copying both from other students' work and from published sources such as books, reports or journal articles.

Exam policy

- If you have to miss an exam, you should make alternative arrangements with the **instructor** before the exam is given.
- If you miss an exam without prior notification, you will receive a zero score for that exam except under truly extenuating circumstances.

Course policy

- **Regular class attendance is required.**
- **If a student routinely misses class, grade reduction will occur.**
- **3 unexcused absences will result in 85 as the best possible score for this course.**

2025 Fall Semester Calendar

四川大學

**National Day holiday &
Mid-Autumn Festival
10/1-10/8
(9/28 and 10/11 makeup days)**

Week 7 Exam 1

Week 13 Exam 2

Week 17 Final Exam

2025—2026 学年（秋季学期）校历

周次		星 期						月份	备 注
		日	一	二	三	四	五		
				2	3	4	5	6	9 月
1	教	7	8	9	10	11	12	13	10 月
2		14	15	16	17	18	19	20	
3		21	22	23	24	25	26	27	
4		28	29	30	1	2	3	4	
5		5	6	7	8	9	10	11	
6		12	13	14	15	16	17	18	
7		19	20	21	22	23	24	25	
8	学	26	27	28	29	30	31	1	11 月
9		2	3	4	5	6	7	8	
10		9	10	11	12	13	14	15	
11		16	17	18	19	20	21	22	
12		23	24	25	26	27	28	29	
13		30	1	2	3	4	5	6	
14		7	8	9	10	11	12	13	
15	周	14	15	16	17	18	19	20	12 月
16		21	22	23	24	25	26	27	
17		28	29	30	31	1	2	3	
18		4	5	6	7	8	9	10	
									1 月

1. 9 月 4-5 日为在校 生 报 到 时 间。
2. 9 月 5-7 日为本科 生 补 缓 考 时 间。
3. 9 月 2-3 日为本科 生 新 生 报 到 时 间，9 月 9-10 日 为 研 究 生 新 生 报 到 时 间。
4. 9 月 8 日 举 行 本 科 生 开 学 典 礼，9 月 12 日 举 行 研 究 生 开 学 典 礼。
5. 第 一 周 起 正 式 行 课。
6. 中 秋 节 10 月 6 日。
7. 春 节 2 月 17 日。
8. 红 色 是 节 假 日，停 课 一 般 不 补。
9. 政 治 学 习、党 团 组 织 生 活 统 一 安 排 在 双 周 星 期 五 下 午 进 行，上 半 段 为 政 治 学 习 时 间，下 半 段 为 党 团 组 织 生 活 时 间。
10. 校 秋 季 田 径 运 动 会 在 10 月 31 日—11 月 1 日 举 行。

Job prospect

<https://www.indeed.com/>

Key words: C++

programmer/developer/software engineer

Job ad 1



BUILD THE FUTURE



C++ Software Engineer, Tesla Bot

[Tesla](#)  6,490 reviews [6,490 reviews](#)

Palo Alto, CA

Full-time

You must create an Indeed account before continuing to the company website to apply

[Apply on company site](#)  

Job details

Here's how the job details align with your job preferences.
Manage job preferences anytime in your [profile](#).

 **Job Type**

Full-time

What to Expect

As a Software Engineer within our robotics teams, you will contribute to one of the most advanced Robotics/AI Platforms in the world.

In this role you will work across multiple functions to ensure seamless integration between various subsystems of the robots including the main computer, sensors, motor controllers, telemetry systems, etc. to enable functionality on existing and new robotics platforms. Your work can cover everything from tuning the robots locomotion stack to building the infrastructure that will scale to the entire robot fleet. You will help implement innovative features that leverage the power of Tesla AI. You will define and maintain content and configurations that support smooth functionality across variations of the robot platform.

What You'll Do

Job ad 2

Software Engineer (C++) – 2024 Grads



Hudson River Trading

New York, NY

\$175,000 - \$250,000 a year

- Their work spans our incredible live trading system and a powerful research environment.
- We have built one of the world's most sophisticated computing...

Posted 8 days ago · More...

Info on the internet

- C++ Wikipedia

<https://en.wikipedia.org/wiki/C%2B%2B>

- Standard C++

- <https://isocpp.org/>

- C++

- <https://cplusplus.com/>

What is C++? (by ChatGPT)

Default (GPT-3.5)

OP

C++



Certainly! C++ is a widely-used programming language that was developed as an extension of the C programming language. It is known for its efficiency, flexibility, and performance, making it a popular choice for various types of software development, including system-level programming, game development, and application development.



Here are some key features and concepts associated with C++:

1. **Object-Oriented Programming (OOP):** C++ supports object-oriented programming, allowing you to create classes and objects. OOP promotes the organization of code into reusable and modular components.
2. **Standard Template Library (STL):** The STL is a collection of C++ template classes to provide general-purpose classes and functions with templates that implement many popular and commonly used algorithms and data structures.
3. **Strong Typing:** C++ is a strongly typed language, which means it enforces type safety during compilation. This helps catch type-related errors at compile time.
4. **Static and Dynamic Memory Allocation:** C++ allows you to allocate memory both statically (at compile time) and dynamically (at runtime) using features like arrays, pointers, and dynamic memory allocation operators like `new` and `delete`.
5. **Pointers and References:** C++ provides direct memory access through pointers and references. Pointers allow you to work with memory addresses directly, while references provide a safer way to access variables without worrying about null pointers.

What is C++? (by ChatGPT)

6. **Operator Overloading:** You can overload operators in C++, enabling you to define custom behaviors for operators when used with user-defined types.
7. **Multiple Inheritance:** C++ supports multiple inheritance, which means a class can inherit properties and behaviors from multiple parent classes.
8. **Exception Handling:** Exception handling allows you to deal with runtime errors and exceptions gracefully, ensuring that your program can recover from unexpected issues.
9. **Standardization:** C++ is standardized by the ISO (International Organization for Standardization). The latest standard as of my knowledge cutoff date is C++17, but newer standards have been released since then.
10. **Compatibility with C:** C++ is backward compatible with C, which means you can use C code in C++ programs, and C++ code can also call C functions directly.

To get started with C++, you would typically need a C++ compiler like GCC (GNU Compiler Collection) or Microsoft Visual C++ for Windows. You would write your C++ code in source files with a `.cpp` extension and then compile it into executable programs that can be run on your computer.

C++ is a versatile language with a steep learning curve, but it offers a lot of power and control to developers, making it suitable for a wide range of applications.

Online Resources

There are many online courses available for learning C++ at various skill levels, from beginner to advanced. Here are some popular platforms and courses where you can learn C++ online:

Online Resources

Udemy:

- "C++ Programming for Beginners" by Udemy: This course is designed for absolute beginners and covers the fundamentals of C++ programming.

edX:

- "C++ Programming Fundamentals" by Microsoft (edX): This course is a part of Microsoft's C++ curriculum and covers the basics of C++ programming.

Online Resources

Coursera:

- "C++ for C Programmers" by Coursera (offered by the University of California, Irvine): This course is suitable for those who are already familiar with programming concepts in C and want to transition to C++. It covers C++ features and object-oriented programming.

Online Resources

Codecademy:

- "Learn C++" by Codecademy: This interactive online platform offers a course that covers the basics of C++.

Pluralsight:

- "C++ Fundamentals Including C++ 17" by Pluralsight: Pluralsight offers a comprehensive course that covers C++ fundamentals, including the features introduced in C++17.

Online Resources

LinkedIn Learning (formerly Lynda.com):

- "C++ Essential Training" by LinkedIn Learning: This course covers the essentials of C++ programming and is suitable for beginners.

CppCon (YouTube):

- CppCon is an annual conference for C++ enthusiasts, and they have a YouTube channel with numerous talks and presentations on various C++ topics. While not a traditional course, it's a great resource for diving deep into C++.

Online Resources

GeeksforGeeks:

- GeeksforGeeks offers a comprehensive set of tutorials and practice problems for C++. It's a valuable resource for self-paced learning and practice.

MIT OpenCourseWare:

- Massachusetts Institute of Technology (MIT) offers a free online course titled "Introduction to C++" as part of their OpenCourseWare initiative. It's more academically oriented.

Books

Consider complementing your learning with C++ textbooks like

- "Accelerated C++" by Andrew Koenig and Barbara E. Moo
- "C++ Primer" by Stanley B. Lippman, Josee Lajoie, and Barbara E. Moo.

How to learn C++

Learning C++ can be an exciting and rewarding experience, but it does require dedication and practice. Here's a step-by-step guide on how to learn C++ effectively:

How to learn C++

Set Clear Goals:

- Determine why you want to learn C++. Are you interested in game development, system programming, web development, or something else? Having a clear goal will help you stay motivated.

How to learn C++

Understand the Basics of Programming:

- If you're completely new to programming, start by learning the fundamentals of programming logic and concepts. Understand variables, data types, control structures (like loops and conditionals), and functions.

How to learn C++

Learn C First (Optional):

- While not necessary, some people find it beneficial to learn the basics of the C programming language before diving into C++. C++ is built upon C, so having a solid understanding of C can provide a strong foundation.

How to learn C++

Select Learning Resources:

- Choose online courses, textbooks, or tutorials that cater to your learning style and goals. The list of online courses and platforms mentioned in the previous response can be a great starting point.

How to learn C++

Practice Regularly:

- Programming is a skill that improves with practice. Write code regularly to reinforce what you learn. Start with simple programs and gradually tackle more complex projects as you gain confidence.

How to learn C++

Master the Basics:

- Focus on understanding variables, data types, operators, control statements (if, for, while), functions, and input/output operations in C++. These are the building blocks of any program.

How to learn C++

Explore Object-Oriented Programming (OOP):

- C++ is known for its support of OOP. Learn about classes, objects, inheritance, polymorphism, and encapsulation. These concepts will allow you to write efficient and modular code.

How to learn C++

Practice with Data Structures and Algorithms:

- Once you are comfortable with the basics, start learning about data structures (e.g., arrays, linked lists, trees) and algorithms. Understanding these concepts is crucial for efficient problem-solving.

How to learn C++

Use Standard Libraries:

- Familiarize yourself with the Standard Template Library (STL), which provides a collection of useful data structures and algorithms. Learning to use these can save you a lot of development time.

How to learn C++

Work on Projects:

Apply your knowledge by working on real projects. Start with small projects and gradually increase the complexity. Building projects helps you solidify your skills and gain practical experience.

Seek Community and Resources:

Join programming forums, communities (like Stack Overflow), and C++ discussion groups. Don't hesitate to ask questions when you're stuck, and contribute to the community as you gain knowledge.

How to learn C++

Read C++ Books (Optional):

Consider reading books specifically focused on C++. Some highly recommended ones include "Accelerated C++" by Andrew Koenig and Barbara E. Moo, and "C++ Primer" by Stanley B. Lippman, Josee Lajoie, and Barbara E. Moo.

Stay Updated:

C++ evolves over time. Stay informed about the latest C++ standards and features. C++17 and newer standards introduce many improvements.

How to learn C++

Practice Problem Solving:

Solve programming challenges on platforms like LeetCode, Codeforces, or HackerRank to sharpen your problem-solving skills.

Build a Portfolio:

Document your projects and code on platforms like GitHub to showcase your skills to potential employers or collaborators.

How to learn C++

Remember that learning programming, especially a language as powerful as C++, takes **time** and **persistence**.

Be patient with yourself, and don't be discouraged by initial challenges. Keep building and experimenting, and you'll gradually become proficient in C++.